AI UX & Data Visualisation Design Principles (CA6002)

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1

1

Chapter 5.2 - HCAI UX Design - Trust

Contents

- Overview
- Components of Trust
- Trust Calibration
- How to Build Trust?
- Who Needs an Explanations
- Design Guidelines for Better AI Explanations
- Design Guidelines for Providing User Control
- Design Guidelines for User Feedback

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2

2

Human-Centred AI User Experience (UX) Design

Overview

- AI systems are **dynamic** and **change over time** as they gather data from the environment and their interactions with users. As a result, the experience of human users with these AI system also needs to adapt to these changes.
- This calls for re-think about how human-centred AI user experience (UX) design is engineered. This chapter focuses on two key aspects in HCAI UX design.
  - **Designing Trust** – There are two essential components needed to build trust for AI systems. The first is **explainability**, which is to ensure that the user understand how the AI system is working for them. The second is the user's sense of **control** over the AI system and how they can make it serves their needs and goals.
  - **Designing Usability** – Usability is how well users can use the AI system to achieve their defined goal. Two aspects of usability design are emphasised, namely the design of intuitive **AI interactions** and effective **error handling**.

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3

Designing Trust

Components of Trust

- Building user's trust with an AI system is like developing a **good relationship**. It takes **repeated experience** with the AI system to build up that positive confidence.
- Some components that contributes user trust in your AI solutions include:
  - **Competence** – this is the ability of your AI solution to do what it **is expected** of it to do. Feature-rich good-looking products that do not meet the **user's need** and **add value** in a meaningful and **easily recognised** manner will not be perceived as competent.
  - **Reliability** – is how **consistently** the AI solution delivers on its designed functionality in a **correct** and **predictable** manner. It may have basic features but it should works all the time and produce expected results. **Predictable interfaces** tend to give users a better experience of reliability.
  - **Benevolence** – is the belief that the AI solution is designed to do good, add value and look after the interest of the user. The outcomes and interactions are **accurate**, **fair**, **unbiased** and respects the user's **autonomy**.

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[1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022
[2] Gale Review, Leveraging LLM for Post-OCR Correction … - https://review.gale.com/2024/09/03/using-large-language-models-for-post-ocr-correction/

4

4

Designing Trust

Trust Calibration

- AI systems are **probabilistic systems** and can sometimes produce erroneous outputs. As such, humans working alongside AI systems can only collaborate effectively when they have the **right level of trust** in the AI solution.
- Users may **distrust** the AI solution when the trust is less then the system's capabilities. This reduces the benefits of AI as users lacking the confidence, **ignore** the AI's recommendation.
- **Overtrust** is when user trust exceeds that of the system's capabilities. Users defer to the AI's recommendation without hesitation when human **judgement** is called for (Note: overtrust can quickly turn into distrust after a bad experience).

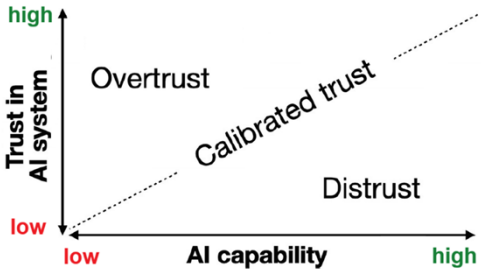


image modified from Akshay Kore

[1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022
[3] US lawyer sorry after ChatGPT creates fake cases in his court filing - https://www.straitstimes.com/world/united-states/us-lawyer-sorry-after-chatgpt-creates-fake-cases-in-his-court-filing

Designing Trust

How to Build Trust?

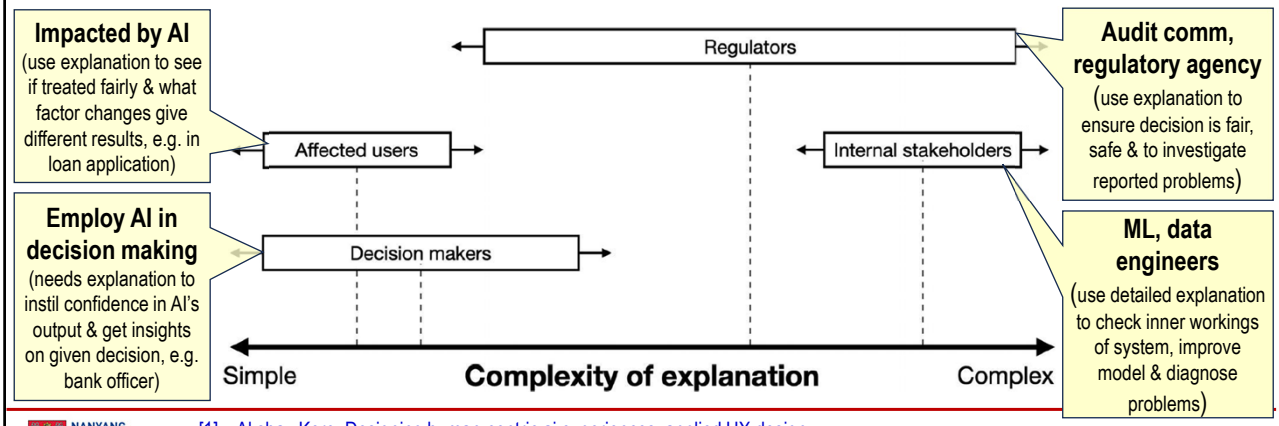
- Building trust is not about being correct all the time. It is about **integrity** in accepting that mistakes happen and **actively correcting** them.
- Users must be assisted to **calibrate their trust** and be able to judge when and how much to trust the **AI's output** and when they need to use their **own judgement**.
- There are two essential parts to building trust for AI systems:
  - **Explainability** – means ensuring users of the AI system understand **how it works** and **how well it works** in order to appropriately calibrate their trust in the AI's output. Explanations must also be optimised for **clarity** and understanding by the target users.
  - **Control** – users will develop more trust if they **feel in control** of their relationship with the AI system. Such control over the AI algorithm (e.g. choosing preference for data used and output types) is likely to improve adoption and give them a sense that the algorithm is superior.

[1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022

Designing Trust

Who Needs an Explanation?

- When designing an AI system, different stakeholder types must be considered as the **complexity of explanation** will vary with their needs and capabilities.



[1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022

Image from Akshay Kore

7

7

Designing AI Explanations

Design Guidelines for Better AI Explanations

- The process of designing AI explanation requires us to build the right **mental model** for users that will help them understand how AI makes decisions and the relationship between inputs and outputs.
- Some design guideline issues that can facilitate the construction of mental models that improve AI explanations are:

1) System capabilities
2) Performance humility
3) Adaptive behaviour
4) Input and output transparency
5) Understandable explanation
6) Confidence level

[1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022
[4] Megan Chan, Mental Models, NN/g, Jan 2024 - https://www.nngroup.com/articles/mental-models/

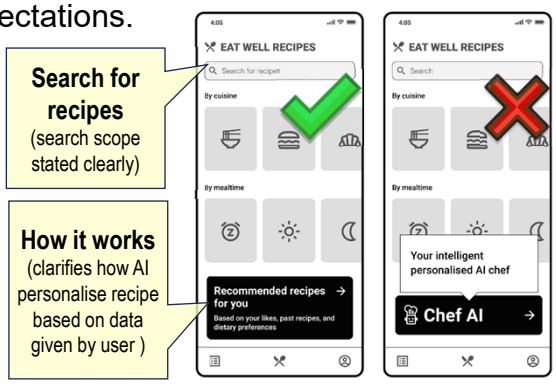
8

8

Designing AI Explanations

Design Guidelines -1) System capabilities

- Helping users understand the capabilities of the AI system by providing **contextual information** on **how the AI systems works** and interacts with the data provided can help build user trust and set correct expectations.
- **Scope of functionality** provided are clearly communicated (e.g. search function that can only works for recipes and nothing else should say so).
- Avoid **open-ended** interaction **prompts** that lack proper guidance (e.g. AI assistants that present prompts like "*Ask me anything*" or "*What would you like me to do?*" may set the wrong expectation about the system's capabilities).



[1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022

Image from Akshay Kore

9

9

Designing AI Explanations

Design Guidelines - 2) Performance humility

- Be realistic about the AI system's performance and **set expectations** at the onset. Inform users what works best for the AI tool based on how it was trained (e.g. a plant recognition systems recognises indoor plants best as it was only trained on indoor plant datasets). Unrealistic expectations degrade user experience.
- AI systems are **probabilistic**, best to avoid using **definitive** language (e.g. instead of "*These are the books you'll like*" use "*We think you'll like these books*").
- **Dynamic** system performance should be communicated in a **timely manner** (e.g. increasing demand or traffic can lead to higher wait times for requested service. Update this info to manage user expectations).

[1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022

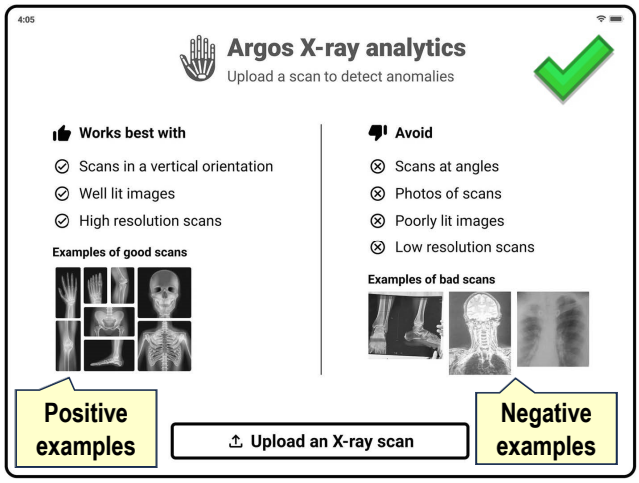
10

10

Designing AI Explanations

Design Guidelines - 2) Performance humility (more)

- AI system's are normally trained and tested with a domain-specific and constrained dataset. It performs well when inputs are of **similar types**.
- **Example-based explanation** can be used to help users develop a clearer mental model of how to get the best performance out of the AI system (e.g. an X-ray image analytics user interface shows sample of image types that the system will perform well on and what it will not).



[1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022

Image from Akshay Kore

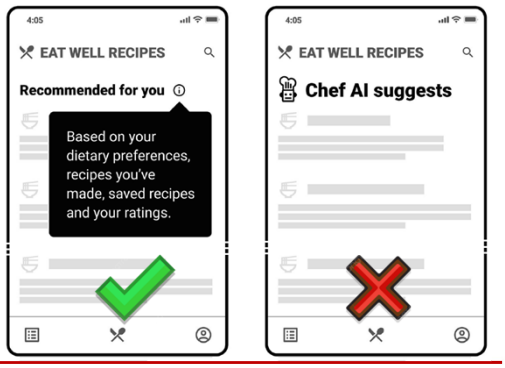
11

11

Designing AI Explanations

Design Guidelines - 3) Adaptive behaviour

- AI systems **adapt** to changing conditions and user behaviour to optimise and personalise their outputs (e.g. Netflix adjusts its recommendation based on your usage patterns and user ratings). Communicate to users how their **inputs influence** the **results** they see and the system adapts and learns over time.
- Results may differ as the AI algorithm optimise using changing data. Trust in the **consistency** of the AI can be maintained by communicating relevant factors affecting the current results (e.g. latest taxi arrival time has increased and is less precise due to today's weather and unpredictable traffic conditions. Reason for increase should be indicated with indication of **confidence level** of arrival time estimate).



[1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022

Image from Akshay Kore

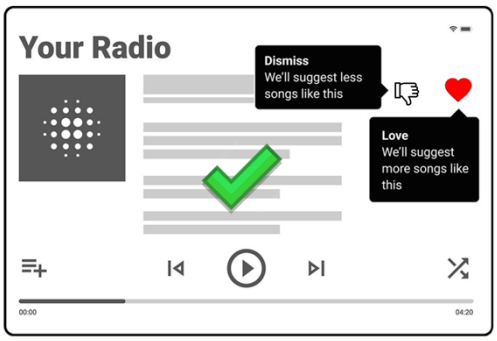
12

12

Designing AI Explanations

Design Guidelines - 3) Adaptive behaviour

- As AI adapts and **improves over time**, so should the user's relationship with the AI systems (e.g. a music recommendation service may not be accurate with new users but over time and usage, the service learns user preference and improves it accuracy and relevance).
- Explaining that the service learns the user's preference over time and improves its prediction or recommendation can help users **calibrate their trust** at different stages of interaction.
- Displaying a measurable level of prediction accuracy (if available) can slowly build this trust through the user's journey over time (e.g. using **confidence level** indicators).



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[1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022

Image adapted from Akshay Kore

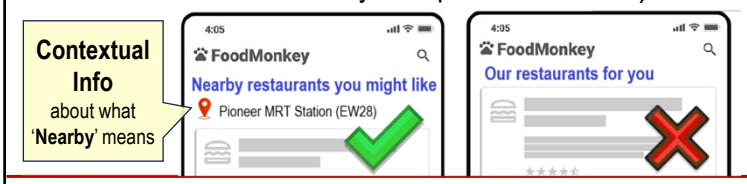
13

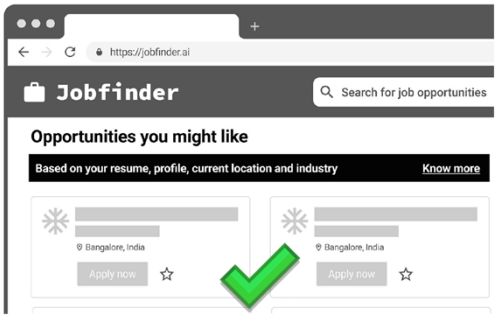
13

Designing AI Explanations

Design Guidelines - 4) Input and Output Transparency

- Transparency is central to earning trust. Users should be made aware of any **data** that is **collected**, **tracked** and **used** by the AI service (e.g. feature like "**Know more**" can be used to provide more details to interested users about the data collected and how it is used).
- Transparency in how the **output is derived** also helps to instil trust as users see how their actions or data affect the recommendation (e.g. shops are recommended because they are within 1 km and not because they are sponsored entities).





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[1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022

image adapted from Akshay Kore

14

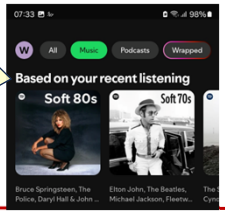
14

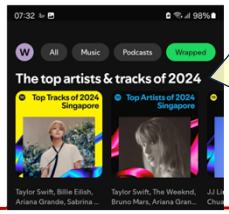
Designing AI Explanations

Design Guidelines - 4) Input Transparency (more)

- To avoid users being surprised to see their personal info being reveal, explain where **their data comes** from (i.e. explain data sources) and how it is being used (e.g. personal assistant booking a cab explains the action is link to user's calendar and preference setting for Grab).
- Explaining the **reach of the data use** helps user understand how the AI derived the recommendations (e.g. data used is personalised to their previous preferences or aggregated across all users).

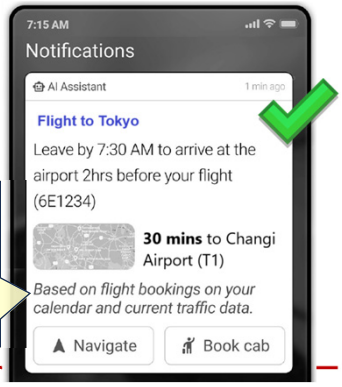
Personal
Based on my listening preferences





Global
Based on preference of all users

Scope of Data
Explain type and range of data used



[1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022

Image adapted from Akshay Kore

15

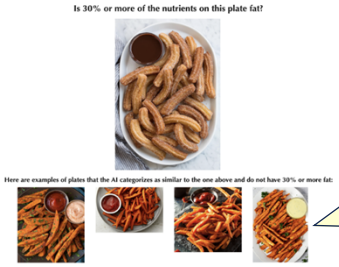
15

Designing AI Explanations

Design Guidelines - 4) Output Transparency (more)

- Explaining the reason behind an AI's output can be aided by the use of example-based explanation. **Examples** help users build **mental models** of the AI's behaviour and aids in their **trust calibration** of the system's current output.
- Displaying the most relevant examples (i.e. specific example-based explanation) to the current interaction helps user judge how much to trust the classification results.

Matches that build trust
The most relevant examples retrieve of salmon looks similar to the given query image, giving confidence that the AI works well



images from Bucinca

Matches that degrades trust
Human judgement easily sees that the churros with chocolate is a different food item to the sweet potato fries with BBQ sauce

[5] Zana Bucinca et. al, Proxy tasks and subjective measures can be misleading in evaluating explainable AI systems, Harvard University, 2020 - https://kgajos.seas.harvard.edu/papers/bucinca20proxy.pdf

16

16

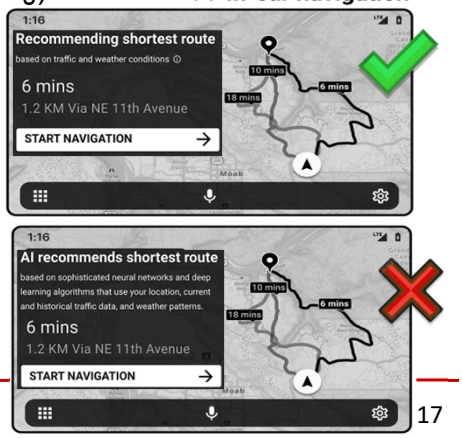
Designing AI Explanations

To be Continued

Design Guidelines - 5) Understandable Explanation

- It can be challenging to explain how the AI system works and detailed explanation can confuse users. So keep it **simple** and **targeted**. (e.g. best not to attempt to explain everything, just aspects that impact user trust and decision-making).
- **Accuracy vs explainability** is an important consideration. Deploying deep learning systems in high stakes use cases can be risky and needs to be properly justified (e.g. medical diagnosis where clear explanations are needed but difficult to extract from the model's output).
- The EU's **GDPR (Article 12)** law governs the way communication must be done (i.e. concise, transparent, intelligible and easily accessible).

In-car navigation



[6] EU GDPR Article 12 - https://gdprhub.eu/Article_12_GDPR

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17

17

Designing AI Explanations

Design Guidelines - 6) Confidence Level

- Confidence level is a **statistical measure** (a value between 0 to 1, where 1 is highest) that indicates how **confident** a probabilistic AI system is about its result.
- Explanation with confidence values can help users assess **how much to trust** the AI's output. But such values are not always **actionable** or **intuitive** to the user, as interpreting such values is challenging (e.g. 0.86 or 0.91. Is a 0.5 difference significant?).
- Improve usability and comprehension of the system must be **tested with users** in order to ensure **their display increase trust** in the AI system rather than degrades.
- Don't show confidence values if they mislead or **create distrust**. Inaccurate high confidence scores may lead to undesirable **overtrust**.
- If confidence-based explanation is to be used, **select the type** most appropriate for the AI application (some of these types include categorical, numeric, N-best results).

[1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022

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18

18

Designing AI Explanations

Design Guidelines - 6) Confidence Level – Categorical

- Instead of numeric values (between 0 to 1), confidence values can be **binned into categories** like *high-medium-low* or *satisfactory-unsatisfactory*, etc to make it easy for users to create a mental model of the level of trust to accord the AI's output.
- It is important to think about issues like the **number** of categories, their respective **thresholds**, and their **associated actions** (e.g. a voice-enabled assistant is calibrated with an appropriate action and response based on the confidence level of the recognised instruction).

High confidence | Medium confidence | Low confidence

[1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022

Designing AI Explanations

Design Guidelines - 6) Confidence Level – Numeric

- Numeric confidence explanation (normally scaled to a percentage) may be **confusing** for the average users as it assumes they know how to **interpret** the probability values (i.e. what they represent) and their **thresholds** (e.g. is 82% confidence level high enough?).
- Users may trust the 82% confidence value for **low stakes** decision (e.g. AI prediction of the song they like) but would be wary for **high stakes** scenarios (e.g. medical diagnosis).
- It is good to provide useful **contextual information** about these numeric values (i.e. what they mean, so such info can help users interpret confidence values and make informed decisions).

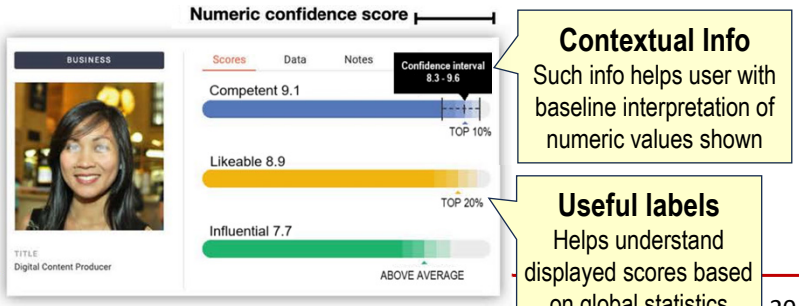
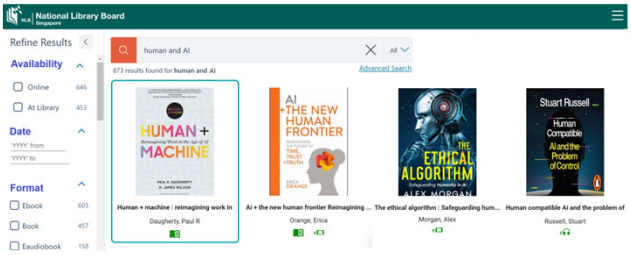


image adapted from Akshay Kore (Photofeeler)

Designing AI Explanations

Design Guidelines - 6) Confidence Level – N-Best Results

- An alternative to explaining using direct confidence levels, the **"n" highest ranked results** (i.e. based on confidence score) of the AI system can be used instead.
- This approach is advantageous in **low-confidence** scenarios or when providing **alternatives** to users is preferrable (i.e. this allows users to rely on their own human **judgement** and **build mental models** of how the AI system relates to the different options shown).



Search results on National Library Board catalogue on 26 Dec 2024

Results displayed in ranked order of highest confidence value (best match).

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[1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022

21

21

Designing User Control

Design Guidelines for Providing User Control

- Another important aspect of designing trust in AI systems is to provide users with **control** over the way the AI **uses their data** and their **interaction with** its **results**.
- The **balance** between the amount of **user control** and **AI automation** depends on the user's context, nature of tasks and the setting in which the AI is deployed.
- **High stake** scenarios (e.g. surgery, criminal justice, finance, etc) will require **more control** by the users. These include greater control and visibility of the algorithmic **decision-making** process, **data** used to train the model and the ability to **monitor** and **correct** the AI's outputs (i.e. via contextual feedback).
- **Low stake** scenarios (e.g. music recommendation, product search, etc) can be more highly automated as inaccurate outputs are **less consequential** to the user.

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[1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022

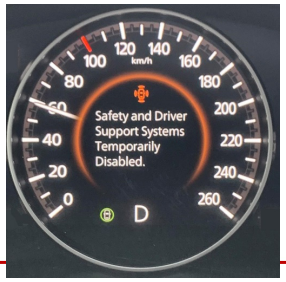
22

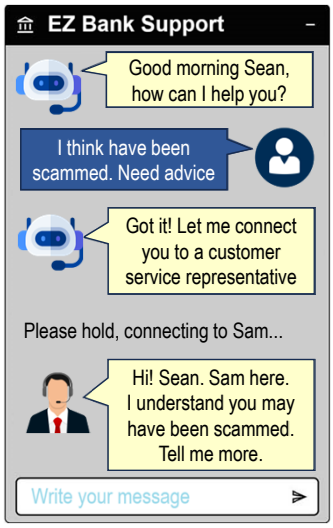
22

Designing User Control

Design Guidelines – Graceful Hand Over

- When the AI system is unable to handle a particular request or situation (e.g. action or query is beyond the current design capabilities of the system) the **AI-to-manual** control transition should be made **easy**, **seamless** and **intuitive** for the user or human agent to take over (e.g. banking chatbot gracefully handing over to a human agent during complicated queries or).
- Ensure users have **awareness** of the AI system's hand off and the **information** they need to take over control (e.g. during poor visibility, users are made aware on the dashboard display that the vehicle safety assistant is no longer working).





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[1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022

23

23

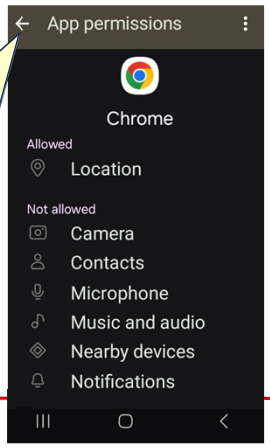
Designing User Control

Design Guidelines – Input Data Control

- Users should have the ability to view, access, edit, share or opt-out from sharing specific aspects of their data with the AI system. **Lack of control** over how users' data is used will **erodes trust** in the system.
- User should be allowed to **enable/disable** the use of **selective data inputs** by the AI system (e.g. enabling location on Chrome but not address contacts).
- Avoid making such controls **too broad** and not selective enough (thus forcing reluctant users to either share all data with the AI service or if not, severely limiting its functionality or worst, not being able to use the service at all).

Permission

Users can selectively enable or disable Chrome's access to different types of data inputs based on how comfortable they are in sharing the data with the app



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[1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022

24

24

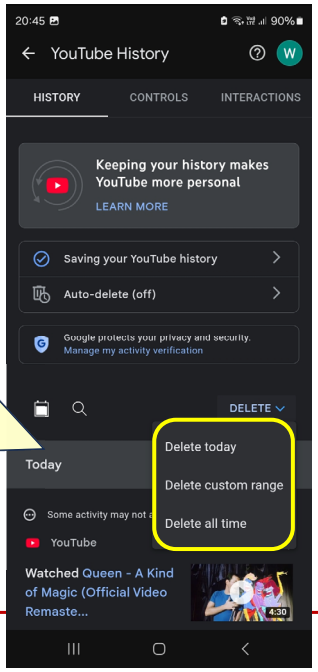
Designing User Control

Design Guidelines – Input Data Control (more)

- Allow users to **delete** or **modify information** that has been **collected** by the AI system (e.g. personalisation process). This may be for privacy reasons or a change in user's personalisation preferences over time (e.g. deleting Youtube history as recommendations are now no longer of one's interest).
- Users should also be allowed to **reset** their **preferences** entirely, especially when the AI system provides a large number of irrelevant suggestions and the best way is to start from scratch (e.g. recommendations from a work-related AI service are no longer relevant due to a recent switch in occupation sector).

Delete History

Users are allowed to delete different extent of their history



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[1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022

25

25

Designing User Control

Design Guidelines – Output Control

- Provide users the **choice of results** (e.g. choose from the 5 top-ranked recommended books displayed) and the empowerment to **adapt** their AI system's **output** to their needs and preferences (e.g. specify the weightage given to a particular matching attribute or period of interest in an information retrieval interaction).

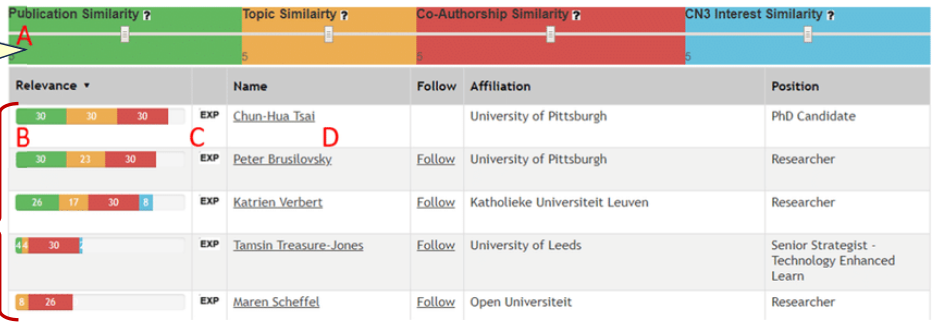
Tsai's Relevance Tuner+

Adapt Output

by adjusting the weights of the different matching attributes

Choice of Results

User can choose what results to view



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[7] C. Tsai, P. Brusilovsky, The effects of controllability and explainability in a social recommender system, User Modeling and User-Adapted Interaction 31(4):1-37, Jul 2021

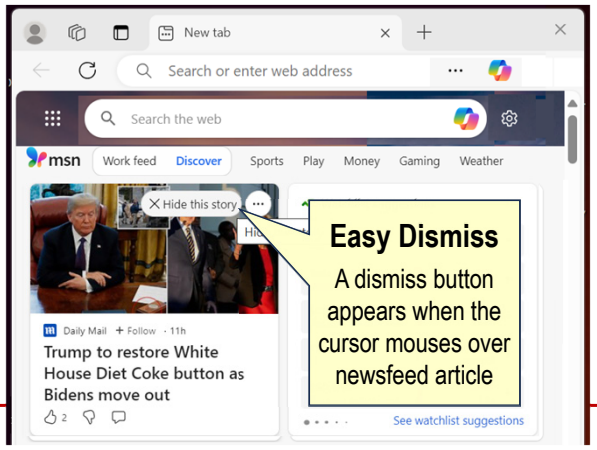
26

26

Designing User Control

Design Guidelines – Output Control (more)

- AI recommender systems are probabilistic and prone to making inaccurate decisions (especially during early interactions when insufficient user behaviour data has been gathered).
- User should be allowed to **efficiently dismiss** any irrelevant or undesirable AI results (e.g. through a side swipe or the click of a dismiss button).
- Besides providing users a greater sense of control over the AI's output, this action also serves to provide **feedback** to the AI algorithms regarding the user's **preferences**.



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[1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022

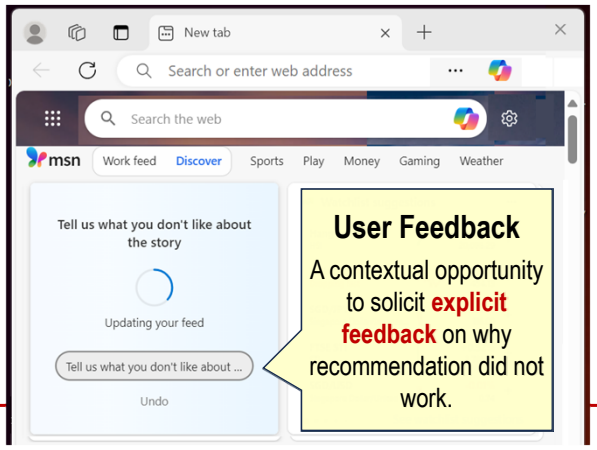
27

27

Designing User Feedback

Design Guidelines for User Feedback

- Feedback is information that comes back as a result of an action. Such **feedback loops** are essential for AI systems that **learn** and **improve** with time.
- Being able to **exploit user feedback** in the interactions is a useful means of improving the AI service and **enhancing** the **user experience**.
- There are essentially two types of feedback:
  - Explicit feedback
  - Implicit feedback



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[1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022

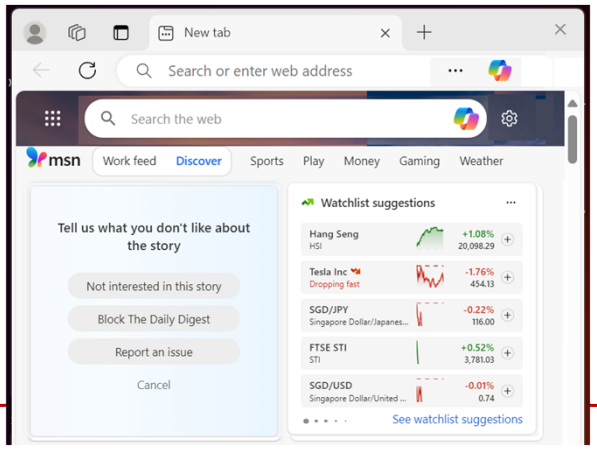
28

28

Designing User Feedback

Design Guidelines – Explicit Feedback

- Explicit feedback is when the users **intentionally give** feedback to improve the AI system (e.g. through surveys, open text fields comments, "thumbs up/down icon" or option clicks).
- Such feedback is useful for getting **unambiguous responses** about the system performance and may imbue users with a **sense of control** over the AI (i.e. "*I can tell* Microsoft Edge *that it is wrong*").
- As they **disrupt** user workflow, be mindful that they are thoughtfully embedded at the **right time** and in the **proper context** (e.g. avoid high-stake situations like driving, surgery, etc).



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[1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022

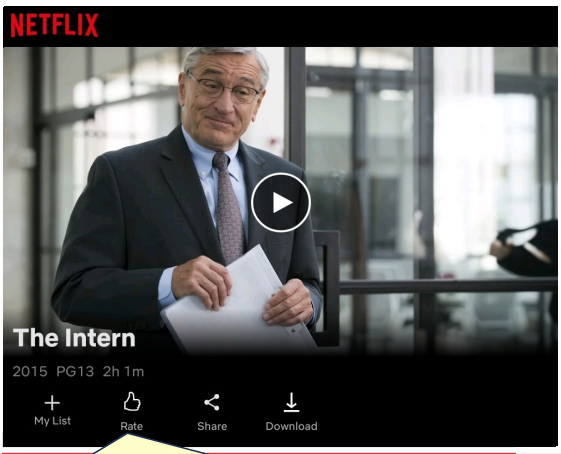
29

29

Designing User Feedback

Design Guidelines – Explicit Feedback (more)

- Feedback response options should be **mutually exclusive** and **understandable** (e.g. thumbs-up vs thumbs down).
- They should be **editable** and **actionable** (e.g. "**Maybe**" is not as it can't use it to improve decision making. "**Others**" option needs further open field text entry. If possible, allow **reversal of action**, users can edit and "**undo**" feedback given earlier).
- Find opportunities to improve trust by **explaining** the **benefits** of feedback and how they **influence** the AI system (e.g. music recommendation services can respond with "*We will show you more songs like this going forward*").



Netflix's compact thumb up/down feedback options

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[8] Ben Shneiderman, Human-Centered AI, Oxford University Press

30

30

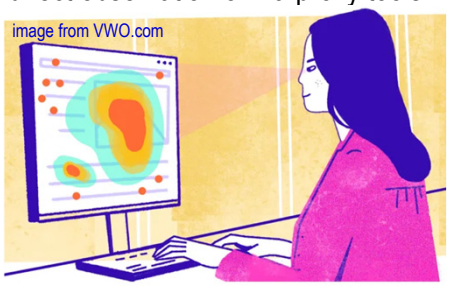
Designing User Feedback

To be Continued

Design Guidelines – Implicit Feedback

- Implicit feedback uses data obtained from user **interactions** (e.g. from product logs that captures in-product metrics) and **behaviour** (e.g. from user direct observation or via proxy tools like eye-trackers, user interviews or social media mentions).
- Reviewing **product logs** helps identify interaction events where users appear **confused** or **error** prone (i.e. events where and when **better explanations** may be needed to help improve user's trust of the system).
- Where possible, **explain** how implicit feedback **is used** within the interaction with the AI system. This can help users calibrate the mental model of the AI's workings (e.g. in monitoring user's movie viewing, subsequent movie recommendations come with the section title "*Because you watched movie M*").

image from VWO.com



Eye-tracker heatmaps show user's dominant visual attention during interactions

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[1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022

31

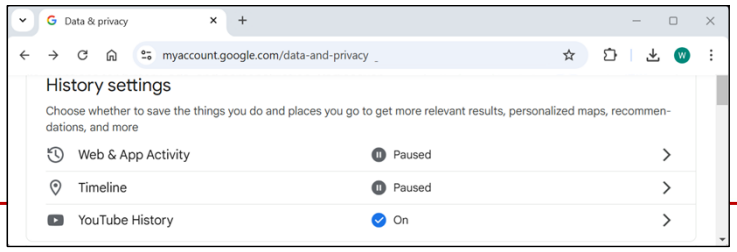
31

Designing User Feedback

Design Guidelines – Implicit Feedback (more)

- Since implicit feedback is usually **obtained covertly** (i.e. without user's awareness), users must **be informed** of what data is collected and how it is used (e.g. Apple Maps privacy policy has a clear and understandable explanation of the data use through implicit feedback).
- **Permission** should be obtained upfront (e.g. in terms of service) and users should be allowed to **opt out** (e.g. this can be done selectively like in the Google account dashboard, where one can opt out of data one is uncomfortable to allow service provider to collect).

Extract from a Google account privacy preference setting web page



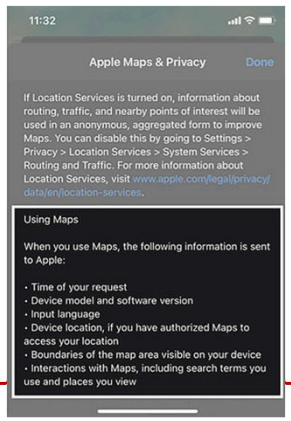


image from Akshay Kore

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32

32

HCAI UX Design – Trust

Summary

- The two design components that significantly influence the **trust** users have for an AI system are **explainability** and a sense of **control** in using the system.
- Designing better explanation requires **understandable communication** and **transparency** about what the **adaptive** AI systems **can do**, the **data** that **it uses** and the quality of **results** that **it produces**.
- Explicit and implicit **feedback** that are **properly collected** from the user can help improve the **performance** of the AI system whilst also building user **trust**.
- Studies have shown that **best user experience** happens when both **explainability** and **controllability** were jointly provided to the user and the these two attributes combined in an **additive** manner to enhance the user's experience of the AI system.

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[7] C. Tsai, P. Brusilovsky, The effects of controllability and explainability in a social recommender system, User Modeling and User-Adapted Interaction 31(4):1-37, Jul 2021

33

33

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[4] Megan Chan, Mental Models, NN/g, Jan 2024 - https://www.nngroup.com/articles/mental-models/

[5] Zana Bucinca et. al, Proxy tasks and subjective measures can be misleading in evaluating explainable AI systems, Harvard University, 2020 - https://kgajos.seas.harvard.edu/papers/bucinca20proxy.pdf

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Note: All online articles were accessed between Nov to Dec 2024

34

34

Chapter 5.2 - HCAI UX Design – Trust

The End